SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – RESEARCH PAPER REVIEW / LITERATURE SURVEY

Course Code: CSA10	Course Title: Software Engineering
Assessment: Research Paper Review / Literature Survey	Weightage: 20% | Total Marks: 25
CO Assessed: CO5	Duration: 60 minutes

CO5 Statement: Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly.

Assigned Problem Statement: Intelligent Disaster Early Warning and Evacuation Management System

Student Name: ________________________________

Register Number: ________________________________

Date: ________________________________

Code of Conduct

I ______________________________________ (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ________________________________

1. Introduction

Disasters — whether triggered by natural hazards such as earthquakes, floods, cyclones, tsunamis and landslides, or by man-made events such as industrial fires and structural failures — continue to cause significant loss of life, infrastructure damage and economic disruption. Traditional early warning and evacuation mechanisms often rely on static thresholds, manual monitoring and fixed evacuation maps, which struggle to adapt to rapidly changing, real-world disaster conditions.

An Intelligent Disaster Early Warning and Evacuation Management System aims to combine sensor networks, the Internet of Things (IoT), edge/fog/cloud computing and Artificial Intelligence (AI) / Machine Learning (ML) techniques to (i) detect and predict hazardous events earlier and more accurately, and (ii) dynamically compute and communicate the safest evacuation routes to affected individuals in real time. This literature survey reviews existing research relevant to this problem statement, compares the approaches used, identifies their limitations, and highlights the research gaps that motivate further work in this area.

1.1 Scope of the Survey

- AI/ML-based hazard prediction and early warning generation (flood, tsunami, fire, slope/landslide, industrial hazards).
- IoT, edge, fog and cloud-based sensing and communication architectures for disaster monitoring.
- Dynamic evacuation route planning and optimization techniques.
- Integration challenges, real-time performance and societal/ethical considerations of deploying such systems.

2. Individual Paper Reviews

Cherian, Jayaraj & Vaidyanathan (2010) — Artificially Intelligent Tsunami Early Warning System

Objective: To design an AI-assisted early warning mechanism for tsunami detection based on seismic and oceanic sensor readings.

Methodology: Rule-based artificial intelligence combined with sensor-driven decision logic to classify seismic events and trigger alerts.

Dataset / Tools: Seismic sensor and tidal-gauge style simulated data.

Key Findings: Demonstrated that automated AI-assisted classification can shorten the decision time between hazard detection and alert issuance compared to manual analysis.

Limitations: Relies on hand-crafted rules and thresholds, limiting adaptability to novel or borderline seismic patterns; lacks integration with an evacuation-guidance component.

Mosavi, Ozturk & Chau (2018) — Flood Prediction Using Machine Learning Models: A Literature Review

Objective: To review and categorize machine-learning approaches used for flood forecasting and prediction.

Methodology: Systematic literature review comparing ANN, SVM, ensemble and hybrid ML models applied to hydrological forecasting.

Dataset / Tools: Secondary data drawn from multiple prior flood-prediction studies and hydrological datasets.

Key Findings: Hybrid and ensemble ML models generally outperform single-model approaches in forecasting accuracy; data quality and resolution strongly affect performance.

Limitations: As a review, it does not propose or validate a new model; highlights that many surveyed studies use region-specific data that does not generalize well.

Al Qundus et al. (2022) — AI/SVM-Based Flood Monitoring and Evacuation Decision Support

Objective: To apply a Support Vector Machine (SVM) model to sensor data to classify flood/no-flood conditions and support evacuation decisions for at-risk rooms/areas.

Methodology: Supervised classification (SVM) on monitored environmental data to produce a binary flood-risk decision that feeds an evacuation trigger.

Dataset / Tools: Simulated/monitored room-level environmental sensor readings.

Key Findings: SVM-based classification enabled timely binary decision-making for evacuation triggering with reasonable accuracy.

Limitations: Binary flood/no-flood classification is coarse and does not provide graded risk levels or route-level guidance for evacuees.

Munawar et al. (2022) — AI, GIS and UAV-Based Safe Evacuation Route Identification

Objective: To protect vulnerable populations during floods by combining AI/ML, Geographic Information Systems (GIS) and Unmanned Aerial Vehicles (UAVs) for path planning.

Methodology: Integration of GIS-based spatial mapping, UAV aerial data collection and AI/ML path-planning algorithms to identify the safest evacuation routes.

Dataset / Tools: GIS spatial layers and UAV-captured aerial imagery.

Key Findings: The combined GIS–UAV–AI pipeline improved identification of safe evacuation paths, particularly for populations vulnerable to sudden flooding.

Limitations: UAV-dependent data collection is weather- and infrastructure-sensitive; system performance was not evaluated at large city-wide scale.

Ahanger et al. (2023) — IoT-Inspired Evacuation Framework with Edge SVM and Cloud Route Optimization

Objective: To build a layered IoT–edge–cloud framework for real-time disaster detection and optimized evacuation-route computation.

Methodology: IoT devices collect real-time environmental data; edge computing applies an SVM classifier to detect emergencies; cloud computing optimizes evacuation paths.

Dataset / Tools: Simulated smart-device sensor streams evaluated through experimental simulation.

Key Findings: Reported strong efficiency gains, including roughly a 5.23-second decision delay, about 96.7% F-measure for emergency detection, and low (about 4.56 mJ) energy consumption.

Limitations: Results are based on simulation rather than a live large-scale deployment; scalability under network congestion was not fully assessed.

Xu et al. (2018) — CLOTHO: Mobile-Cloud IoT Evacuation Planning with Artificial Potential Field

Objective: To provide crowd evacuation planning through a mobile cloud computing platform.

Methodology: IoT-based data collection combined with an Artificial Potential Field (APF) algorithm to compute repulsive/attractive forces guiding crowd movement away from hazards toward exits.

Dataset / Tools: Mobile cloud platform with simulated crowd-movement and sensor data.

Key Findings: APF-based planning produced smoother, congestion-aware crowd evacuation paths compared to static shortest-path routing.

Limitations: APF-based methods can suffer from local-minima trapping in complex building layouts; mobile-cloud latency can affect real-time responsiveness.

Bhatia et al. (2025) — IoT–Fog–Cloud (IFC) Assisted Industrial Disaster Evacuation Framework

Objective: To design a three-layer IoT-Fog-Cloud architecture for real-time monitoring and optimized evacuation in industrial disaster settings.

Methodology: Fog-layer pre-processing of spatiotemporal sensor data (to reduce latency/energy use) followed by cloud-based evacuation-route computation and rescue-team coordination.

Dataset / Tools: Spatiotemporal sensor readings simulated for an industrial-site scenario.

Key Findings: The layered fog–cloud architecture reduced latency and energy consumption compared to cloud-only processing, while improving optimized route identification for rescue teams.

Limitations: Framework validated primarily through simulation for an industrial context; adaptability to other disaster types (e.g., wildfire, earthquake) not directly tested.

Researchers using ANN-based route risk scoring (2019 study base) — SmartEscape: An IoT-Based Intelligent Fire Evacuation System

Objective: To identify the shortest and safest fire-evacuation route by assessing dynamic, real-time surrounding conditions.

Methodology: An Artificial Neural Network (ANN) computes a personal usage risk score for each link/path segment using environmental sensor data and evacuee characteristics; risky links are excluded before route calculation.

Dataset / Tools: Building-installed environmental sensors (smoke, heat, occupancy) and evacuee smartphone data.

Key Findings: The ANN-based risk-scoring approach achieved a reported 98.1% accuracy in predicting link-level risk, guiding evacuees with real-time vocal and visual instructions.

Limitations: High accuracy was reported under tested conditions but depends heavily on dense sensor coverage, which raises deployment cost for existing buildings.

ScienceDirect study (2024) — DWM-Evac: Dynamic Graph Neural Network for IoT-Based Fire Evacuation

Objective: To overcome delays in traditional IoT-based dynamic path adjustment during large-building (shopping mall) fire emergencies.

Methodology: Combines Dynamic Graph Neural Networks (DGNN) with IoT evacuee-tracking data to continuously restructure the evacuation-path graph as fire conditions change.

Dataset / Tools: IoT sensor and evacuee-tracking data for a simulated large-shopping-mall fire scenario.

Key Findings: The DGNN-based model processed large-scale IoT data more efficiently than traditional static or rule-based path-adjustment methods, improving real-time responsiveness.

Limitations: Model complexity and computational cost of continuous graph updates may limit deployment on lower-power edge devices.

3. Comparative Literature Survey Table

Author / Year	Research Problem	Methodology	Dataset / Tools	Key Findings	Limitations	Research Gap
Cherian et al. (2010)	Tsunami early detection	Rule-based AI on seismic/tidal sensor data	Simulated seismic data	Faster alert decision vs. manual analysis	Static rules; no evacuation link	No adaptive learning or route guidance
Mosavi et al. (2018)	Flood prediction (review)	Comparative review of ANN/SVM/ensemble ML	Secondary hydrological datasets	Hybrid models outperform single models	No new model proposed; region-specific data	Lack of generalizable cross-region models
Al Qundus et al. (2022)	Flood monitoring for evacuation	SVM binary classification	Simulated room sensor data	Timely flood/no-flood decisions	Coarse binary output only	No graded risk or route-level detail
Munawar et al. (2022)	Safe evacuation routing (flood)	GIS + UAV + AI path planning	GIS layers, UAV imagery	Improved safe-route identification	Weather-dependent UAV data	Not validated at city scale
Ahanger et al. (2023)	Real-time evacuation optimization	IoT + edge SVM + cloud optimization	Simulated IoT sensor streams	Low latency (~5.2s), high F-measure (~96.7%)	Simulation only; network-congestion effects untested	Needs live large-scale deployment validation
Xu et al. (2018) – CLOTHO	Crowd evacuation planning	Mobile-cloud IoT + Artificial Potential Field	Mobile cloud platform, simulated crowd data	Smoother, congestion-aware paths	Local-minima trapping; cloud latency	Needs hybrid APF + learning-based correction
Bhatia et al. (2025)	Industrial disaster evacuation	3-layer IoT-Fog-Cloud architecture	Simulated spatiotemporal industrial data	Lower latency/energy vs. cloud-only	Tested for industrial context only	Cross-disaster-type adaptability unexplored
SmartEscape (2019 base)	Fire evacuation routing	ANN-based per-link risk scoring	Building sensors + smartphone data	98.1% link-risk prediction accuracy	Needs dense sensor coverage (cost)	Low-cost sparse-sensor alternative needed
DWM-Evac (2024)	Large-building fire evacuation	Dynamic Graph Neural Network + IoT	IoT tracking data, mall fire simulation	Efficient real-time path restructuring	High computation cost on edge devices	Lightweight DGNN variant for edge deployment

4. Comparative Analysis of Existing Approaches

Across the reviewed literature, the surveyed approaches can be grouped into two broad, often loosely connected, categories: (a) hazard detection / early-warning generation, and (b) evacuation-route planning and guidance. Comparing them on common parameters highlights clear trade-offs:

4.1 Accuracy and Detection Performance

ANN and DGNN-based approaches (e.g., SmartEscape, DWM-Evac) report the highest reported accuracy figures (around 98% for link-risk prediction), but require dense sensor instrumentation. SVM-based approaches (Al Qundus et al.; Ahanger et al.) achieve strong classification performance with comparatively lower computational overhead, making them more deployable on edge devices.

4.2 Real-Time Responsiveness

Edge- and fog-layer pre-processing (Ahanger et al.; Bhatia et al.) consistently reduces latency compared to sending raw data directly to the cloud, which matters greatly for time-critical evacuation decisions. Purely cloud-centric or rule-based systems (Cherian et al.) show higher decision latency.

4.3 Scalability and Deployment Cost

UAV- and dense-sensor-dependent methods (Munawar et al.; SmartEscape) provide rich, accurate data but raise deployment cost and weather-related reliability concerns. Lighter IoT-edge-cloud frameworks scale more easily but may sacrifice some prediction granularity.

4.4 Integration of Warning and Evacuation

Most early-warning-focused papers (tsunami, flood-classification, review studies) stop at hazard detection and do not extend into route-level evacuation guidance. Conversely, most evacuation-focused papers (CLOTHO, DWM-Evac, IFC framework) assume a hazard has already been detected and focus purely on path optimization. Very few reviewed works tightly couple both stages into a single, end-to-end intelligent system.

5. Identified Research Gaps

- Limited end-to-end integration: most systems address either early detection or evacuation routing, but not both within a unified, tightly-coupled pipeline.
- Heavy reliance on simulated data: the majority of reviewed studies validate performance through simulation rather than large-scale, real-world field deployment.
- Scalability across disaster types: frameworks are often designed and validated for a single hazard (e.g., fire, flood, industrial incident) with little evidence of cross-hazard generalizability.
- Edge-device computational constraints: high-accuracy models (ANN, DGNN) are computationally demanding, limiting deployment on low-power edge/IoT hardware in resource-constrained regions.
- Inclusion of vulnerable populations: few systems explicitly model the needs of elderly, disabled, or otherwise vulnerable evacuees in route computation.
- Geographic research imbalance: existing case studies concentrate in well-resourced regions (North America, Europe, East Asia), leaving disaster-prone but resource-constrained regions under-studied.

6. Future Research Directions

- Design lightweight, edge-deployable hybrid models (e.g., pruned/quantized DGNN or ANN variants) that retain high accuracy while running on low-power IoT/edge hardware.
- Develop unified architectures that couple hazard-detection models directly with dynamic evacuation-route optimization in a single real-time pipeline.
- Validate proposed frameworks through real-world pilot deployments and field trials rather than simulation alone, to assess robustness under actual network and environmental conditions.
- Incorporate accessibility-aware and vulnerability-aware routing that accounts for mobility-impaired, elderly, and other at-risk evacuees.
- Extend research coverage to resource-constrained and disaster-prone regions currently underrepresented in the literature.

- Explore federated and privacy-preserving learning approaches so multiple buildings, cities, or agencies can collaboratively improve prediction models without sharing raw sensor data.

7. Conclusion

The reviewed literature demonstrates substantial progress in applying AI/ML and IoT-edge-cloud architectures to disaster early warning and evacuation management, with individual studies achieving strong results in hazard classification, route optimization and latency reduction. However, the survey also reveals a clear gap between hazard-detection research and evacuation-planning research, a heavy dependence on simulated rather than field-validated data, and limited attention to vulnerable populations and cross-hazard generalizability. These gaps directly motivate the design of an integrated, lightweight, and inclusive Intelligent Disaster Early Warning and Evacuation Management System that couples real-time detection with adaptive, accessibility-aware evacuation guidance.

8. References

(IEEE referencing style — renumber/reformat according to your instructor's preferred citation style if different.)

[1] Cherian, C. M., Jayaraj, N., & Vaidyanathan, S. G. (2010). Artificially Intelligent Tsunami Early Warning System. In Proceedings of the 12th International Conference on Computer Modelling and Simulation (UKSIM), 39–44.

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[6] Xu, Q., et al. (2018). CLOTHO: A Mobile Cloud Computing Platform for Crowd Evacuation Planning Using an Artificial Potential Field Algorithm. [As cited in: arXiv:2502.07787].

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[9] DWM-Evac: Fire Emergency Management of Large Shopping Malls — IoT-Based Evacuee Tracking and Dynamic Path Optimization Using Dynamic Graph Neural Networks (2024). ScienceDirect.

[10] Additional background: The Role of Artificial Intelligence for Early Warning Systems: Status, Applicability, Guardrails and Ways Forward. PMC12637242.